Tectonophysics, 186 (1991) 269–278
Elsevier Science Publishers B.V., Amsterdam

Paleomagnetism and tectonics of the Jura arcuate mountain belt in France and Switzerland

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(Received February 18, 1990; accepted May 1, 1990)

ABSTRACT

Gehring, A.U., Keller, P. and Heller, F., 1991. Paleomagnetism and tectonics of the Jura arcuate mountain belt in France and Switzerland. *Tectonophysics*, 186: 269–278.

Goethite and hematite in ferriferous oolitic beds of Callovian age from the Jura mountains (Switzerland, France) carry either pre- and/or post-tectonic magnetization. The frequent pre-tectonic origin of goethite magnetization indicates a temperature range during formation of the arcuate Jura mountain belt below the goethite Néel temperature of about 100 °C. The scatter of the pre-tectonic paleomagnetic directions ($D = 11.5°$ E, $I = 55.5°$; $\alpha_{95} = 4.7$) which reside both in goethite and hematite, provides strong evidence that the arcuate mountain belt was shaped without significant rotation. The paleomagnetic results support tectonic thin-skinned models for the formation of the Jura mountain belt.

Introduction

The Jura mountains (Folded Jura) in northern Switzerland and eastern France formed as a classical foreland belt in the Late Miocene during the last Alpine compression. The arcuate shape of the mountain belt is bound towards the north and west by the Oligocene Rhine–Bresse graben system. Oligocene normal faulting affected the sediments as well as the basement of the younger Folded Jura. Folds and thrusts are restricted to the Mesozoic cover which is separated by Triassic evaporites from the Paleozoic basement. The overthrusts and folds strike E–W in the eastern part and bend in an almost continuous manner to a N–S trend in the western regions (Fig. 1). The Tabular Jura is the autochthonous foreland to the north and has been mainly affected by Oligocene normal faulting.

Thick- and thin-skinned tectonic models have been developed in order to explain the formation of the arcuate shape of the Jura mountains (see Laubscher, 1980). The thick-skinned models postulate sinistral strike-slip faults in the basement and en-échelon folds in the sediments above (Pavoni, 1961, Wegmann, 1963). A consequence of these thick-skinned models is *anticlockwise* rotation of the sedimentary cover in the Folded Jura. Pavoni's (1961) model requires an anticlockwise rotation of about 15 degrees, whereas 20 degrees of rotation are necessary to satisfy Wegmann's (1963) reconstruction.

In the thin-skinned models, the Mesozoic cover of the Folded Jura is detached above the Triassic evaporites as a consequence of crustal shortening in the Alps (Buxtorf, 1907; Laubscher, 1961, 1980). The divergent thrust and fold directions in the Jura mountain arc are explained by differential block movements in the detached sediments. With the direction of movement controlled by the orientation of strike-slip faults. The geometry of the block boundaries is strongly influenced by the Oligocene Rhine–Bresse graben system. The thin-skinned models suggest *clockwise* rotation of the Folded Jura during Alpine deformation of about 8 degrees (Laubscher, 1965, 1980, 1986).

Paleomagnetic data from the Jura mountains are very scarce and restricted to the central and

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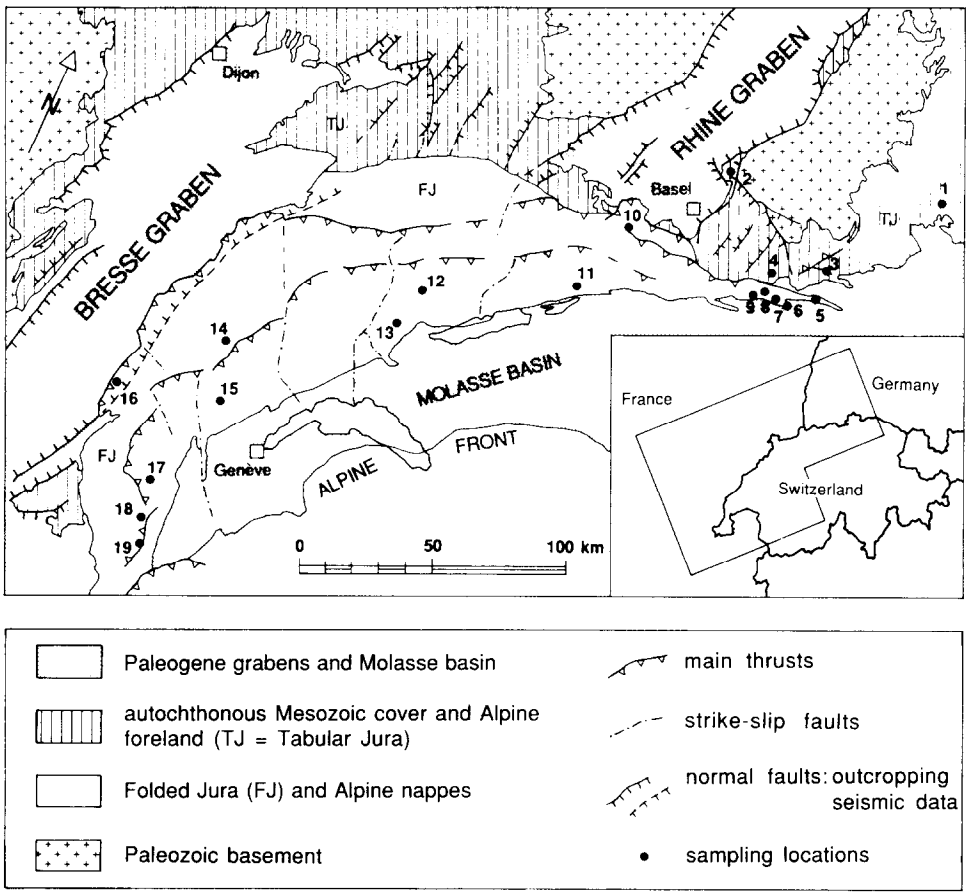


Fig. 1. Tectonic map of the Jura mountains and adjacent areas. For sampling locations see Table 1.

eastern part. Johnson et al. (1984) established primary magnetization in Late Jurassic diagenetically unaltered blue micritic limestones and determined a Jurassic paleopole for the whole Jura mountain range which is in close agreement with other paleopoles from the stable areas of Europe (Heller, 1977; Kadzialko-Hofmokl and Kruczik, 1987). The data from Johnson et al. (1984), which cover only a relatively small proportion of the Folded Jura, have been taken by Eldredge et al. (1985) to demonstrate that the arcuate shape of the belt was not caused by oroclinal bending. Gehring and Heller (1989) showed that goethite and/or hematite in ferriferous beds of Middle Jurassic age within the Folded Jura can carry a magnetization of pre-tectonic origin. The present study aims at enlarging the paleomagnetic data base by using the widely distributed exposures of these ferriferous beds. Thus paleomagnetic constraints may be set up in order to test the tectonic models which either propose anticlockwise rotation of the Folded Jura by 15 to 20 degrees, or alternatively involve minor clockwise rotation of the mountain arc.

Sampling, methods and material

Oriented samples from ferriferous oolitic beds of Callovian age (Middle Jurassic) were collected from nineteen localities distributed across the arcuate belt and the northern, non-folded Tabular Jura (Fig. 1, Table 1). These ferriferous beds represent highly condensed sequences, usually less than 1 m thick, containing hardgrounds and a rich ammonite fauna (Mangold, 1970; Gehring, 1989). The ooids sit in a marly or calcareous matrix and consist mainly of berthierine (7 Å trioctahedral serpentine), or goethite (α-FeOOH) and less frequently hematite (α-Fe_2O_3). The paleomagnetic samples were drilled, and the 2.5 cm diameter

TABLE 1

Sampling localities and mean paleomagnetic directions

Site	Locality	Demagnet. range (°C)	R/N	Dec/Inc (BTC)	α_{95}	Dec/Inc (ATC)	α_{95}	Magnetization type
1	Blumberg	50–100	0/20	356.0/ 44.0	3.0	356.0/ 44.0	3.0	C
2	Kandern	no pre-tectonic direction						B
3	Mandach	50–150	0/18	0.5/ 31.3	1.6	23.7/ 65.7	1.8	C
4	Herznach	100–600	2/29	359.0/ 50.2	3.7	359.0/ 50.2	3.7	D
5	Unterehrendingen	50–125	0/19	239.6/ 71.4	7.6	2.7/ 48.9	6.0	C
6	Holderbank	50–125	0/15	191.3/ 2.8	9.0	14.1/ 52.1	5.5	C
7	Wildegg	50–125	0/13	10.3/ 38.6	3.4	24.0/ 61.1	3.4	C
8	Gatter	50–500	3/10	16.6/ 22.2	11.2	16.6/ 57.4	6.6	A
9	Schellenbrücke	100–400	0/ 5	12.2/ 24.9	11.5	14.5/ 62.9	11.5	D
10	Liesberg	25–500	0/26	228.0/ 65.8	5.5	20.0/ 44.6	4.6	A
11	Reuchenette	50–650	5/55	7.5/ 58.7	2.3	11.2/ 59.1	2.3	A
12	S. Sulpice	150–400	0/14	352.8/ 44.8	5.3	36.1/ 61.7	5.2	D
13	Baumes	150–300	0/ 4	351.0/ 35.9	8.9	24.3/ 62.6	10.9	D
14	Sauget	no pre-tectonic direction						B
15	S. Claude	50–125	6/ 6	48.4/ 55.7	5.0	3.9/ 63.7	5.0	C
16	Selignac	50–100	0/12	341.8/ 42.6	8.5	353.2/ 45.1	6.9	C
17	Arvière	50–100	11/ 0	134.9/ − 44.2	4.9	185.8/ − 53.4	4.6	C
18	Lucey	125–650	0/25	329.1/ 51.2	2.1	18.5/ 46.8	2.2	D
19	Monthoux	no pre-tectonic direction						B

Locations 1-4 are in Tabular Jura, all other sites lie in the Folded Jura. R/N is the number of samples with reversed polarity (R) and normal polarity (N), respectively. Geological detail description of sites to be found in Gygi (1969), Mangold (1970), Gehring (1989). Dec/Inc with BTC and ATC denote site mean declination and inclination before and after tectonic dip correction, respectively. α_{95} is Fisher's (1953) 95% circle of confidence. Magnetization types A–D according to description in the text.

cores orientated, using a magnetic compass and an inclinometer, in the field. The natural remanent magnetization (NRM) was measured with an ScT three-axes cryogenic magnetometer. Progressive thermal demagnetization was performed using a Schonstedt oven. The characteristic NRM components were evaluated using linear regression techniques aided by visual inspection of linear segments of the NRM vectors during demagnetization. Isothermal remanent magnetization (IRM) acquisition in fields up to 5.0 Tesla followed by thermal demagnetization was used to identify the different ferromagnetic minerals by means of coercivity and blocking temperature (T_B) spectrum analyses.

Paleomagnetic results

NRM intensities measured in the oolitic samples vary over a wide range between 0.09 mAm^{-1} and 116 mA m^{-1}. Thermal demagnetization reveals that the NRM consists of at least two components, a strong component which often contains both normal and reversed polarity generally carries about 90% of the initial NRM and decays rapidly below 125 °C, and a second component which trends linearly towards the origin of orthogonal vector plots (Zijderveld, 1967) between 150 °C and 680 °C (Fig. 2). Upon heating, samples containing mainly berthierine ooids often show a strong increase in the low field magnetic susceptibility at about 450 °C. At this temperature, berthierine breaks down and magnetite is formed (Gehring and Karthein, 1990). The viscous magnetization in this newly formed ferromagnetic mineral phase often prevents accurate determination of the high temperature NRM component in the samples.

IRM acquisition of unheated oolitic samples is controlled by high coercivity minerals which are not completely saturated in a 5 Tesla field (Fig. 3a). Thermal demagnetization of the IRM shows two major kinks around 100 °C and between 625 °C and 680 °C (Fig. 3b). The unblocking tem-

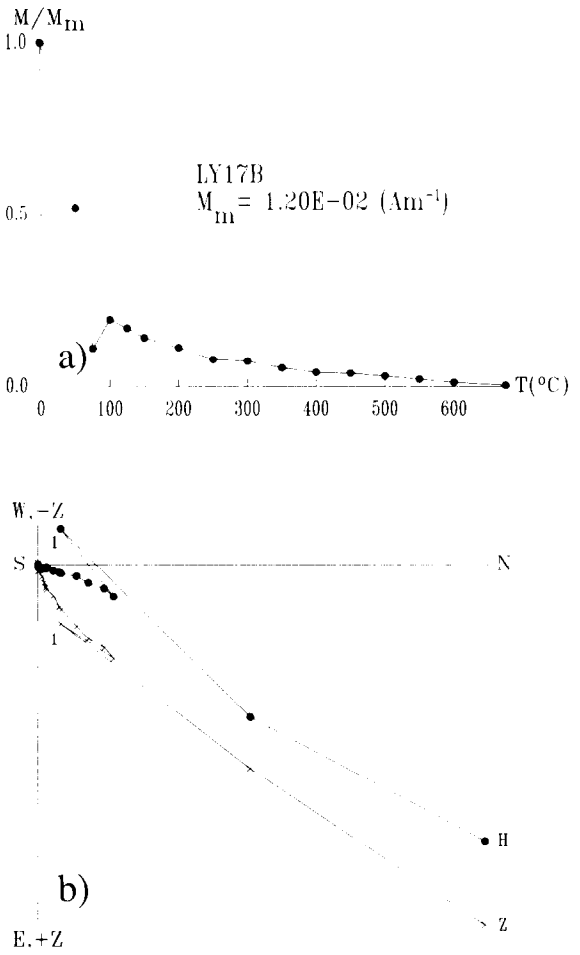


Fig. 2. Intensity (a) and vectorial destruction (b) of NRM during thermal demagnetization of an oolitic sample. H and Z denote horizontal and vertical components, respectively.

perature at about 100 °C is typical of goethite and the higher one of hematite (Lowrie and Heller, 1982). Goethite and hematite have also been identified by XRD methods (Gehring, 1989) as the only ferric compounds in the oolitic beds. The magnetizations at the nineteen sampling sites are carried by two ferric mineral phases, goethite and hematite. The magnetization histories can be divided into two groups, which are dependant upon whether the goethite and hematite components are parallel or different. Each group can be further subdivided, depending on the pre-tectonic or post-tectonic origin of the magnetization components. The resulting four types of magnetization are illustrated in the following examples (Figs. 4 and 5). In the first group, the goethite and hematite components are parallel (Fig. 4). This indicates a single component of magnetization, representing the characteristic remanent magnetization (ChRM).

Type A: At Liesberg (Fig. 1, Table 1: Site 10) a single component magnetization is detected between 25 °C and 500 °C. Its direction dips steeply towards the southwest before tectonic dip correction (BTC). After tectonic correction (ATC) a northerly, moderately steep mean direction is observed which is in good agreement with the Jurassic field configuration of stable Europe (Irving, 1977). Since the BTC direction is far off any Jurassic or post-Jurassic direction to be expected in stable Europe, but the ATC direction is close (Fig. 4d), a pre-tectonic origin for the ChRM of type *A* is postulated.

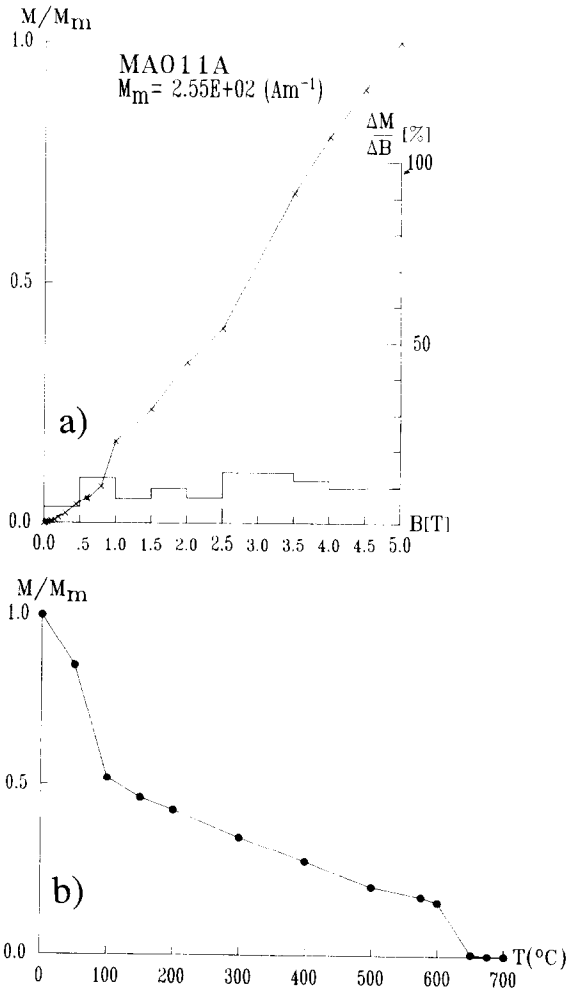


Fig. 3. (a) IRM acquisition and resulting coercivity spectra and (b) IRM thermal demagnetization of an oolitic sample.

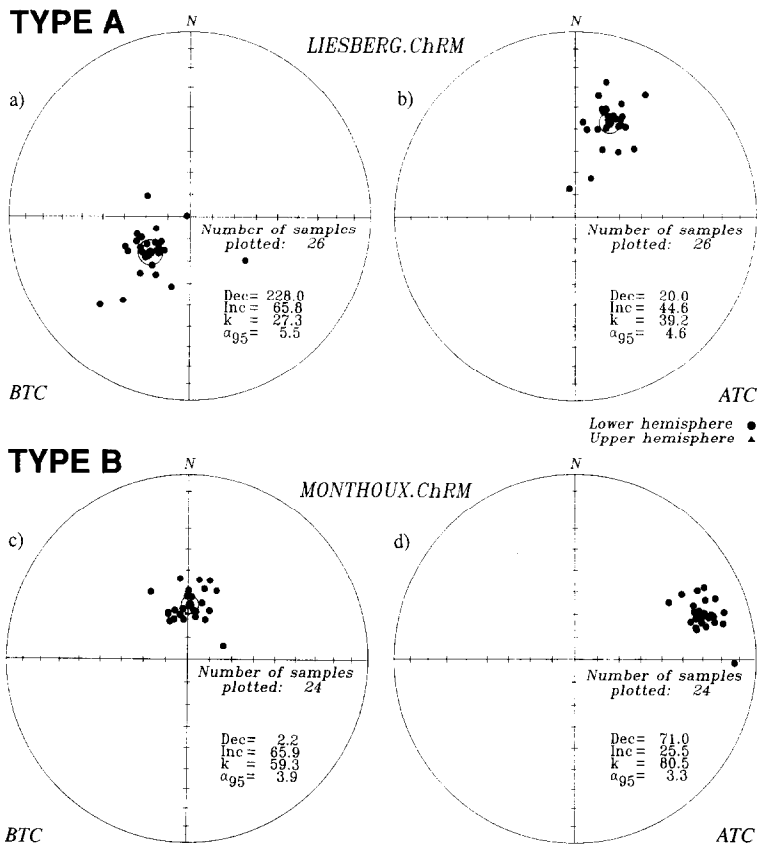


Fig. 4. Equal area projection of the ChRM of the iron oolite at Liesberg (site 10; magnetization type *A*) and Monthoux (site 19; magnetization type *B*).(a) and (c) Before tectonic dip correction (BTC). (b) and (d) After tectonic dip correction (ATC). Dots = lower hemisphere; triangles = upper hemisphere.

Type B: At Monthoux (Fig. 1, Table 1: Site 19) a well defined BTC single component ChRM of normal polarity, stable between 25° C and 600° C, parallels the present earth field direction, whereas the corresponding ATC direction plots far away from any field direction expected since Jurassic times (Fig. 4a,c). Therefore the ChRM is interpreted as being of post-tectonic origin. Type *B* magnetizations are discarded from the tectonic discussion.

The magnetization history of the majority of the oolitic beds of the 19 sites can be considered to belong to the second group in which the two ferric phases are magnetized at different times with respect to the main tectonic event in the Jura mountains.

Type C: At Mandach (Fig. 1, Table 1: Site 3) a low temperature unblocking component (T_B < 150° C) residing in goethite shows a BTC declination due north and an inclination of about 30° (Fig. 5a) which is too shallow for a Jurassic or post-Jurassic field configuration. After tectonic correction a mean declination of 24° E and an inclination of 66° results. A pre-folding origin is interferred for this magnetization (Fig. 5b). In contrast, the NRM component carried by hematite (T_B > 150° C) is oriented very closely towards the present or late Tertiary field before tectonic dip correction (Fig. 5c). The over-steep ATC mean direction with an enlarged circle of confidence strongly suggests a post-tectonic magnetization (Fig. 5d). Therefore this hematite magnetization will be excluded from further discussion.

Type D: At Lucey (Fig. 1, Table 1: Site 18) the BTC goethite NRM (T_B < 100° C) shows normal and reversed polarities with a normal mean declination of 12° E and an inclination of 58° (Fig. 5e). The abnormally shallow ATC directions (mean: D = 44° E and I = 31°, Fig. 5f) are interpreted as indications of two antiparallel magneti-

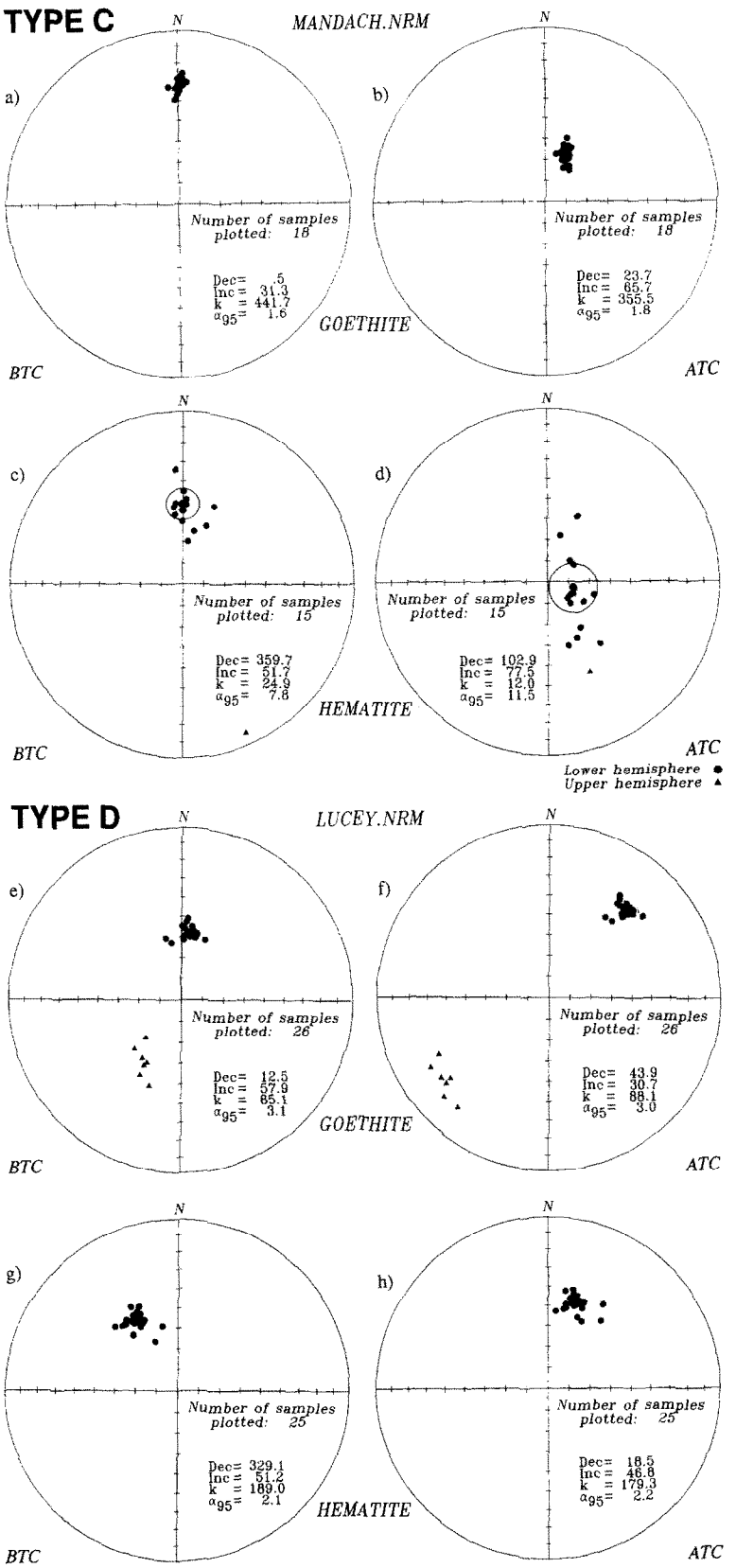


Fig. 5. Equal area projection of the NRM. (a) and (b) after tectonic correction of goethite components; (c) before and (d) after tectonic correction of hematite components; in the iron oolite at Mandach (site 3; magnetization type *C*), (e) before and (f) after tectonic correction of goethite components; (g) before and (h) after tectonic correction of hematite components; from Lucey (site 18; magnetization type *D*). Dots = lower, hemisphere, triangles = upper hemisphere.

zations of post-tectonic origin in goethite. In contrast, the hematite NRM (Fig. 5g,h) has a BTC mean declination of 329° E and an inclination of 51° and an ATC declination of 18° E with an inclination of 47°. From the comparison of the characteristic directions of the hematite and the goethite it is concluded that the hematite components are of pre-tectonic origin.

The comparison of all the paleomagnetic data shows that three of the nineteen sites under investigation (Table 1) contain goethite and hematite carrying a post-tectonic magnetization only. The mean direction of the pre-tectonic NRM components at the remaining sixteen sites has a declination of 11.5° E and an inclination of 55.5° with a circle of confidence (α_{95}) of 4.7°.

Discussion

The magnetization of the oolitic beds being carried by hematite and/or goethite has been formed at different times with respect to the main Jura tectonic events. The frequent occurrence of goethite with a pre-tectonic magnetization origin indicates a temperature regime below the goethite Néel temperature of about 100° C during the formation of the Jura mountain arc. This supports the earlier findings of Gehring and Heller (1989) who estimated a temperature range of less than 90° C for the eastern Jura.

The paleomagnetic data of the oolitic beds containing ferric mineral phases with pre-tectonic magnetization suggest neither systematic change nor regional dependency of the site mean directions over the whole Jura arc (Fig. 1, Table 1). Instead, the minor scatter of the site mean values (Table 1) may be explained by local thrust sheet rotation. An example for such a variation can be demonstrated at Holderbank and Wildegg (Sites 6, 7: Fig. 1, Table 1). Here in the eastern part of the Jura belt, the ATC site mean declinations are different by about 10 degrees. Leaving aside the question of statistical error, this can be interpreted as the result of thrust sheet rotation since a similar amount of local rotation has been postulated by Bitterli (1988) in the eastern Jura mountains based on three-dimensional material balancing methods.

Positive response to a fold test which takes into account all sites in the Folded Jura including those of Johnson et al. (1984, p. 483) is demonstrated in Fig. 6. The ATC mean direction of the Folded Jura sites plots within the cone of confidence of the mean direction of the Tabular Jura (Fig. 7). The field of expected declinations for the thick-skinned models (Fig. 7, field no. 2), however, is clearly outside of both circles of confidence.

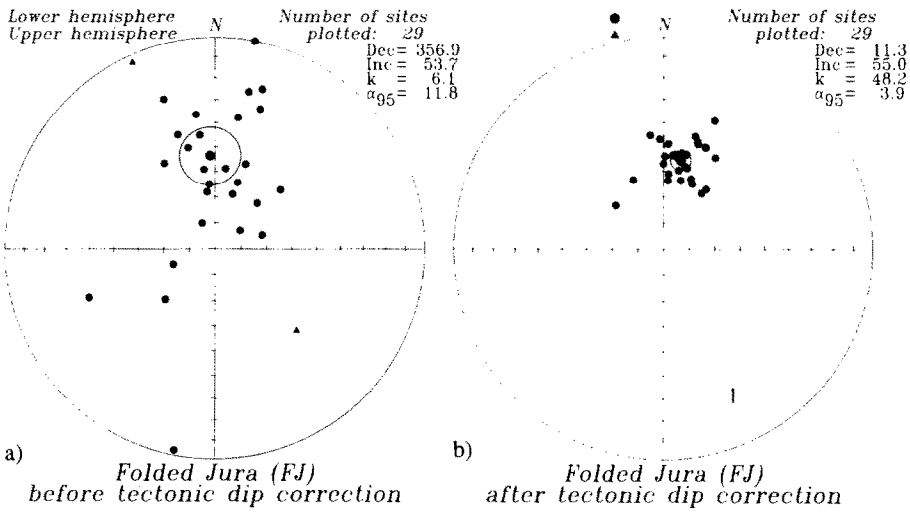


Fig. 6. Equal-area projection of all available Folded Jura sites (data from this study and from Johnson et al., 1984) (a) before tectonic dip correction and (b) after tectonic dip correction. Notice the difference in Fisher's (1953) k values. Dots = lower hemisphere; triangles = upper hemisphere.

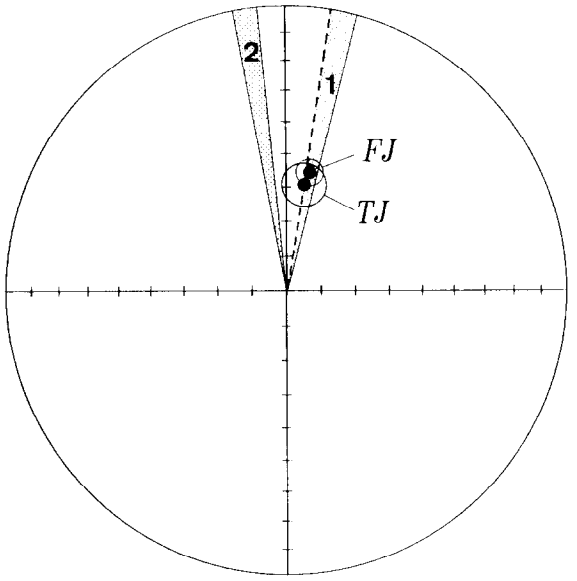


Fig. 7. Regional mean directions of the Folded Jura (*FJ*; Fig. 6b) and of the Tabular Jura (*TJ*; mean declination = 7.8° E, mean inclination = 57.7°, $\alpha_{95} = 6.6°$). Ten sites from this study (Table 1) and Johnson et al. (1984). In addition the expected declination fields due to tectonic rotation are indicated for the thin-skinned (field no. 1) and the thick-skinned models (field no. 2).

Paleomagnetic evidence thus contradicts all models which require anticlockwise rotations by postulating basement movements with resulting en-échelon folding of the Mesozoic cover in the Folded Jura.

The thin-skinned models, which explain the Jura formation by translation of the Mesozoic cover, are the most suitable for a structural interpretation of the paleomagnetic data. A small clockwise rotation of about 8 degrees, predicted by Laubscher (1965), is within the confidence limits of our data (Fig. 7, field no. 1). This rotation is derived from an increasing shortening in the Folded Jura towards west. Alternatively, this increase in shortening can be explained by strike-slip faults. However, new investigations in the Swiss Molasse basin to the south of the Jura range show shortening indicated by overthrusts and duplex-structures in the underlying Mesozoic sediments (NAGRA, 1988). This shortening has to be taken into account in tectonic reconstructions of the Jura mountain arc. Thus a deformation model without regional rotation seems to be possible and equally well supported by the paleomagnetic data.

The lack of significant rotation in the paleomagnetic data, and the continuation of the Mesozoic fold structures beneath the Molasse basin towards the Alps, support thin-skinned models which result in the construction of the Jura mountains as a décollement structure. The arcuate shape of the Jura mountains and its increasing shortening towards the west is explained by differential block translation (Fig. 8). This schematically displayed block outline considers some of the generalities of the Jura fault and thrust geometry (cf. Fig. 1). Under unidirectional (from SE to NW) Alpine compression the different translation direction of the individual blocks was dictated by

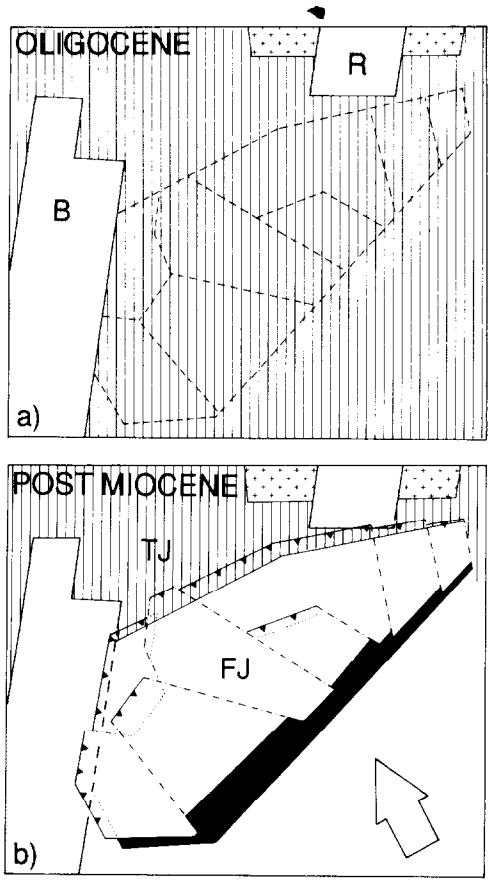


Fig. 8. Schematic tectonic model of the Jura mountain arc based on strike-slip faulting and divergent block translations (modified after Laubscher, 1980). (a) Late Oligocene situation; *B* = Bresse Graben; *R* = Rhine Graben; vertical ruling = undeformed sedimentary cover of the Alpine foreland; dashed lines = hypothetical block structures. (b) Post-Miocene situation. Arrow = estimated direction of the Alpine compression; dash-dotted lines = strike-slipe faults; lines with triangles = overthrusts; dotted lines = overthrust block boundaries.

the orientation of the strike-slip faults in the Mesozoic cover. These faults were either inherited structures from the Rhine–Bresse Graben system or new structures formed during Alpine compression (see Laubscher, 1980, 1986). In the eastern Jura the block translation was mainly towards the north leading to E–W striking compressive structures. In the western Jura, however, the strike-slip faults caused westwards block translation which resulted in N–S striking folds and thrusts (Fig. 8). In this way the N–S striking Bresse Graben edge was overthrust towards the west northwest by the Mesozoic Jura sediments.

Conclusion

The paleomagnetic data of the middle Jurassic oolitic beds indicate that the temperature range during the formation of the Jura mountain belt was below the goethite Néel temperature of about 100 °C. Therefore both major ferromagnetic minerals (hematite and goethite) in the ferriferous beds often carry a pre-tectonic magnetization suitable for tectonic reconstructions. The internal consistency of the paleomagnetic directions at the sixteen new sites of this study and the earlier sites measured by Johnson et al. (1984) which are distributed over the whole Jura mountain range, and the close agreement with the European reference direction suggest that the arcuate shape is a consequence of translations of the Mesozoic sedimentary cover which was detached from the basement along Triassic evaporites. This evidence supports Laubscher's (1961, 1980) model, which proposes that the structure of the Jura mountain belt is a result of compression due to shortening of the basement in the Alps without significant rotation.

Acknowledgements

The authors are grateful to W. Gruber for the field assistence and to H. Laubscher and W. Lowrie for critical reading of the manuscript. This research was supported by the Swiss National Science Foundation, project No. 2.514-0.87.

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